import

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

اختيار الملفات world_happin…_2026 (3).csv
world_happiness_2026 (3).csv(text/csv) - 9842 bytes, last modified: 29100% - 2026/4/ done
Saving world_happiness_2026 (3).csv to world_happiness_2026 (3).csv

```
df=pd.read_csv(r"world_happiness_2026 (3).csv")
df
```

	rank	country	region	score	gdp_per_capita	social_support	healthy_life_expectancy	freedom	genero
0	1	Finland	Western Europe	7.764	1.892	1.581	0.952	0.670	0
1	2	Iceland	Western Europe	7.701	1.874	1.611	0.959	0.662	0
2	3	Denmark	Western Europe	7.688	1.887	1.562	0.948	0.665	0
3	4	Costa Rica	Latin America and Caribbean	7.439	1.254	1.421	0.891	0.631	0
4	5	Sweden	Western Europe	7.401	1.878	1.501	0.952	0.658	0
...	...	...	...	...	...	...	...	...	
125	126	Central African Republic	Sub-Saharan Africa	3.441	0.171	0.691	0.341	0.261	0
126	127	South Sudan	Sub-Saharan Africa	3.421	0.361	0.691	0.341	0.221	0
127	128	Sierra Leone	Sub-Saharan Africa	3.401	0.351	0.741	0.371	0.311	0
128	129	Congo DR	Sub-Saharan Africa	2.761	0.191	0.601	0.311	0.201	0
129	130	Afghanistan	South Asia	1.446	0.241	0.341	0.361	0.071	0

130 rows × 10 columns

Next steps: Generate code with df New interactive sheet

describe

```
df.tail(10)
```

	rank	country	region	score	gdp_per_capita	social_support	healthy_life_expectancy	freedom	genero
120	121	Yemen	Middle East and North Africa	3.541	0.411	0.831	0.571	0.241	0
121	122	Haiti	Latin America and Caribbean	3.521	0.361	0.801	0.571	0.261	0
122	123	Lesotho	Sub-Saharan Africa	3.501	0.541	0.791	0.341	0.361	0
123	124	Burundi	Sub-Saharan Africa	3.481	0.191	0.771	0.451	0.301	0
124	125	DR Congo	Sub-Saharan Africa	3.461	0.281	0.781	0.411	0.311	0
125	126	Central African Republic	Sub-Saharan Africa	3.441	0.171	0.691	0.341	0.261	0
126	127	South Sudan	Sub-Saharan Africa	3.421	0.361	0.691	0.341	0.221	0
127	128	Sierra Leone	Sub-Saharan Africa	3.401	0.351	0.741	0.371	0.311	0
128	129	Congo DR	Sub-Saharan Africa	2.761	0.191	0.601	0.311	0.201	0
129	130	Afghanistan	South Asia	1.446	0.241	0.341	0.361	0.071	0

```
df.head(10)
```

	rank	country	region	score	gdp_per_capita	social_support	healthy_life_expectancy	freedom	generosi
0	1	Finland	Western Europe	7.764	1.892	1.581	0.952	0.670	0.1
1	2	Iceland	Western Europe	7.701	1.874	1.611	0.959	0.662	0.2
2	3	Denmark	Western Europe	7.688	1.887	1.562	0.948	0.665	0.2
3	4	Costa Rica	Latin America and Caribbean	7.439	1.254	1.421	0.891	0.631	0.1
4	5	Sweden	Western Europe	7.401	1.878	1.501	0.952	0.658	0.2
5	6	Norway	Western Europe	7.392	1.959	1.553	0.952	0.661	0.2
6	7	Netherlands	Western Europe	7.319	1.877	1.488	0.942	0.642	0.2
7	8	Israel	Middle East and North Africa	7.234	1.803	1.521	0.905	0.599	0.1
8	9	Luxembourg	Western Europe	7.228	2.096	1.502	0.940	0.649	0.1
9	10	Switzerland	Western Europe	7.201	1.993	1.551	0.961	0.647	0.2

Next steps: Generate code with df New interactive sheet

```
df.describe()
```

	rank	score	gdp_per_capita	social_support	healthy_life_expectancy	freedom	generosity	c
count	130.000000	130.000000	130.000000	130.000000	130.000000	130.000000	130.000000	1
mean	65.500000	5.454185	1.135446	1.176754	0.732015	0.483162	0.164823	
std	37.671829	1.255681	0.559534	0.259117	0.190572	0.118050	0.055821	
min	1.000000	1.446000	0.171000	0.341000	0.311000	0.071000	0.061000	
25%	33.250000	4.136000	0.671000	0.941000	0.563500	0.393500	0.131000	
50%	65.500000	5.581000	1.121000	1.241000	0.791000	0.491000	0.161000	
75%	97.750000	6.396000	1.631000	1.378500	0.891000	0.571000	0.201000	
max	130.000000	7.764000	2.121000	1.611000	0.961000	0.671000	0.301000	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 130 entries, 0 to 129
Data columns (total 10 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   rank                    130 non-null    int64
 1   country                 130 non-null    object
 2   region                  130 non-null    object
 3   score                   130 non-null    float64
 4   gdp_per_capita          130 non-null    float64
 5   social_support          130 non-null    float64
```

```
 6   healthy_life_expectancy  130 non-null    float64
 7   freedom                  130 non-null    float64
 8   generosity               130 non-null    float64
 9   corruption               130 non-null    float64
dtypes: float64(7), int64(1), object(2)
memory usage: 10.3+ KB
```

```
df.pivot_table(index=['region'],values='rank',aggfunc='mean').sort_values('rank',ascending=False)
```

region	rank
Sub-Saharan Africa	104.186047
South Asia	80.000000
Middle East and North Africa	75.071429
Southeast Asia	60.444444
Commonwealth of Independent States	56.600000
Latin America and Caribbean	51.333333
East Asia	44.000000
Central and Eastern Europe	28.454545
Western Europe	17.333333
North America and ANZ	15.500000

```
df.pivot_table(index=['region'],values='score',aggfunc='mean').sort_values('score',ascending=False)
```

region	score
Western Europe	6.987167
North America and ANZ	6.969000
Central and Eastern Europe	6.551273
East Asia	6.099200
Latin America and Caribbean	5.874933
Commonwealth of Independent States	5.757000
Southeast Asia	5.694333
Middle East and North Africa	5.356929
South Asia	4.816833
Sub-Saharan Africa	4.204256

```
cols = ['score' ,'gdp_per_capita',  'social_support',   'healthy_life_expectancy',  'freedom',  'generosity',

result = {}

for col in cols:
    highest = df.loc[df[col].idxmax(), 'country']
    lowest = df.loc[df[col].idxmin(), 'country']

    result[col] = {
        'highest': highest,
        'lowest': lowest
    }
```

```
result
summary = pd.DataFrame({
    'highest_country': [df.loc[df[col].idxmax(), 'country'] for col in cols],
    'lowest_country': [df.loc[df[col].idxmin(), 'country'] for col in cols]
}, index=cols)

summary
```

Show hidden output

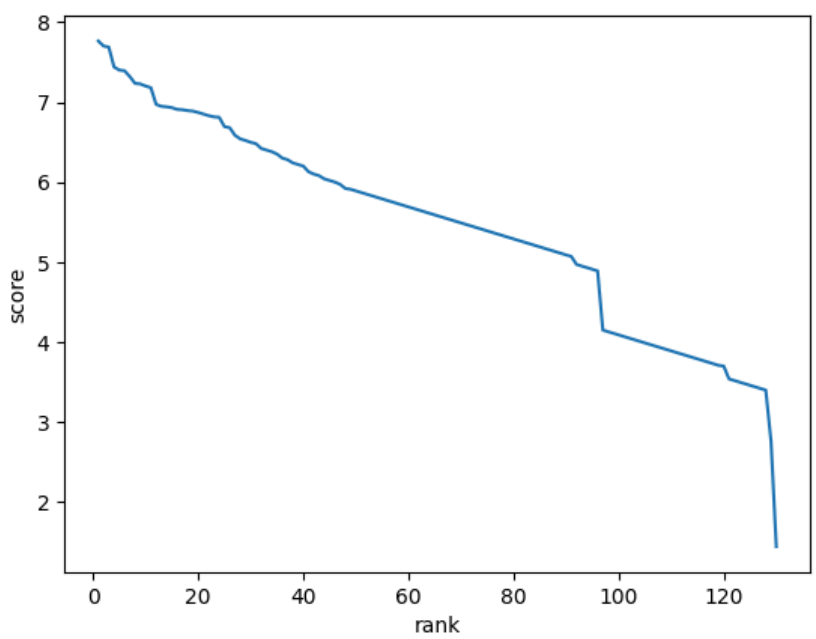
Next steps: Generate code with summary New interactive sheet

```
lowest = df.loc[df['rank'].idxmin()]
highest = df.loc[df['rank'].idxmax()]
resulte=pd.concat([lowest, highest], axis=1)
resulte
```

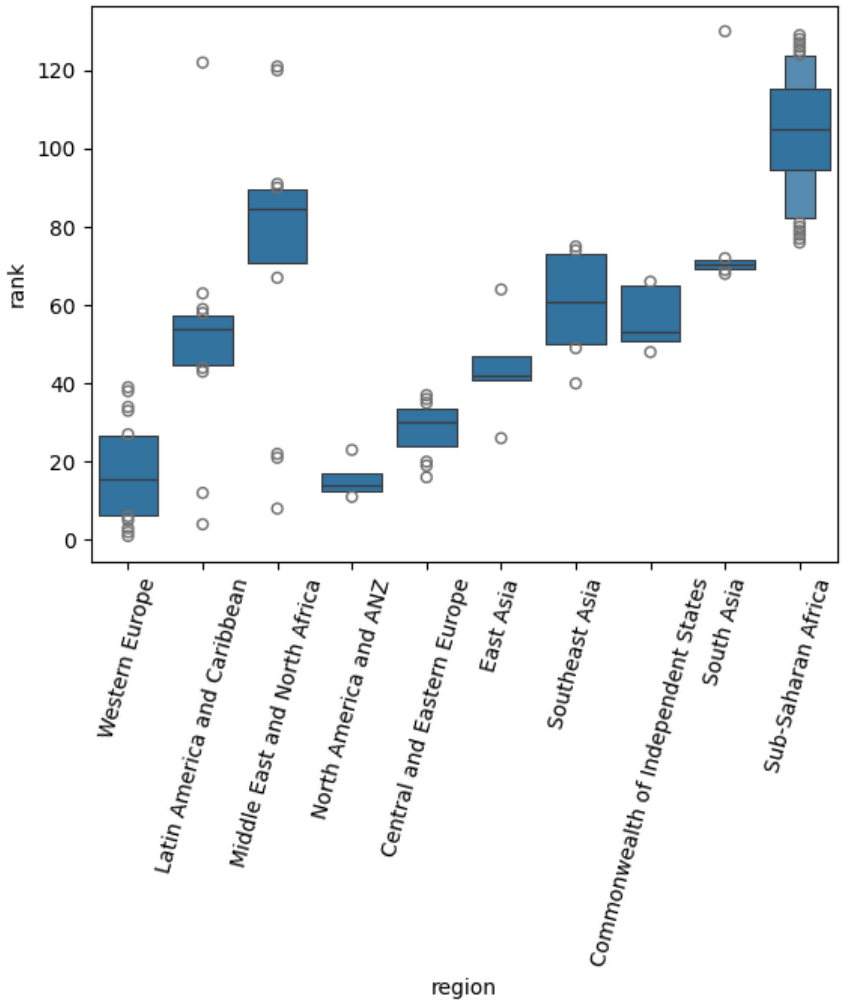
	0	129
rank	1	130
country	Finland	Afghanistan
region	Western Europe	South Asia
score	7.764	1.446
gdp_per_capita	1.892	0.241
social_support	1.581	0.341
healthy_life_expectancy	0.952	0.361
freedom	0.67	0.071
generosity	0.186	0.071
corruption	0.498	0.011

Next steps: Generate code with resulte New interactive sheet

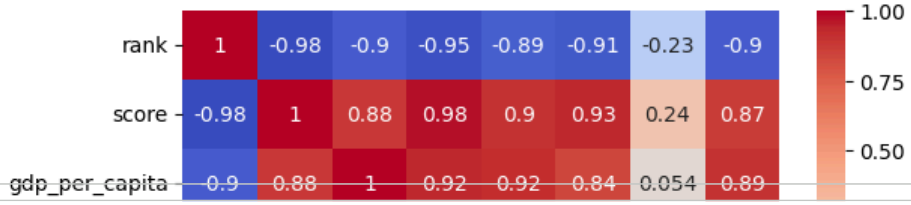
```
sns.lineplot(data=df ,x='rank' ,y='score')
plt.show()
```

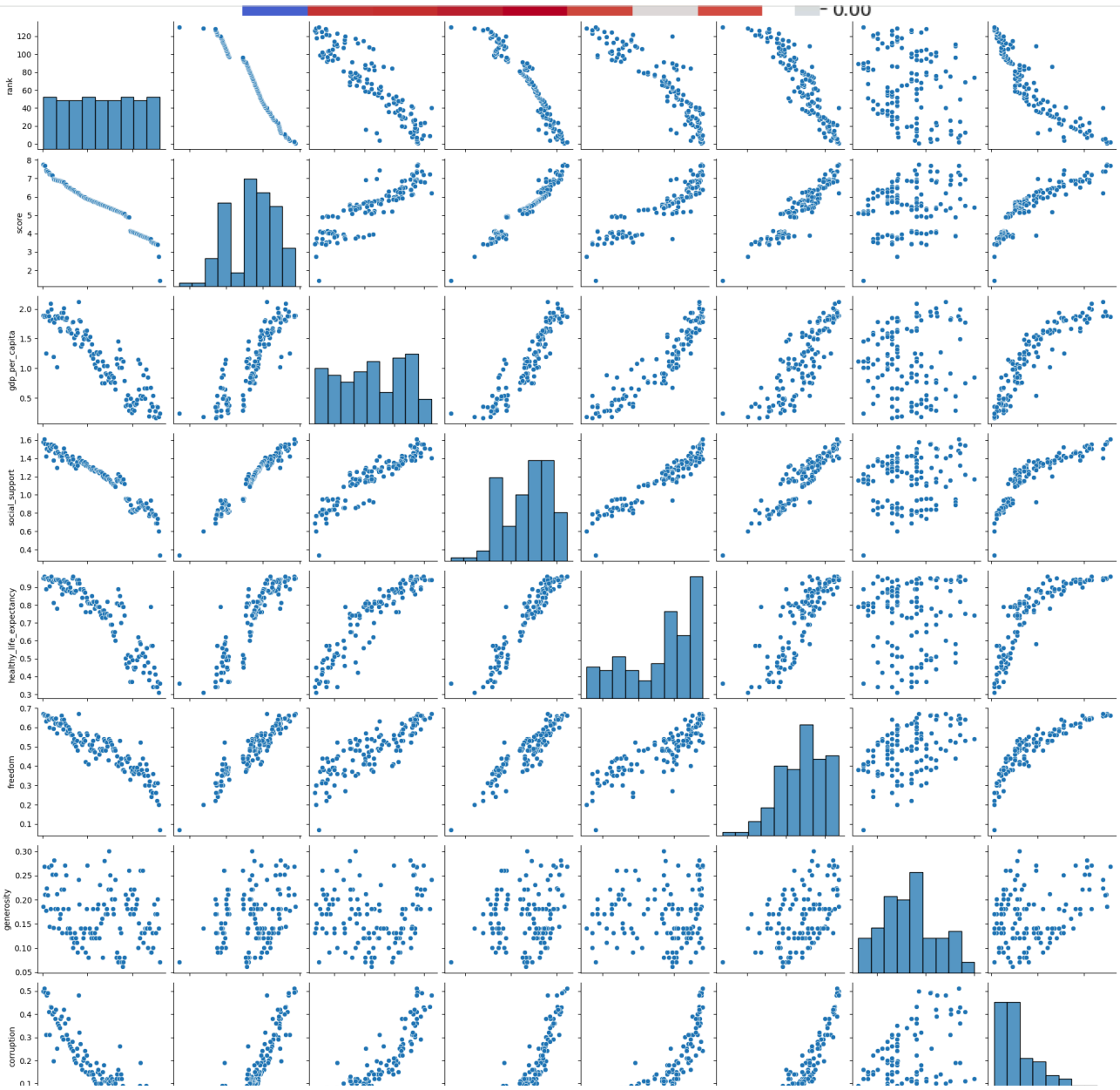


```
sns.boxenplot(data=df,x="region",y="rank")
plt.xticks(rotation=75)
plt.show()
```




```
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True,cmap="coolwarm")
plt.show()
```




```
sns.pairplot(data=df)
plt.show()
```